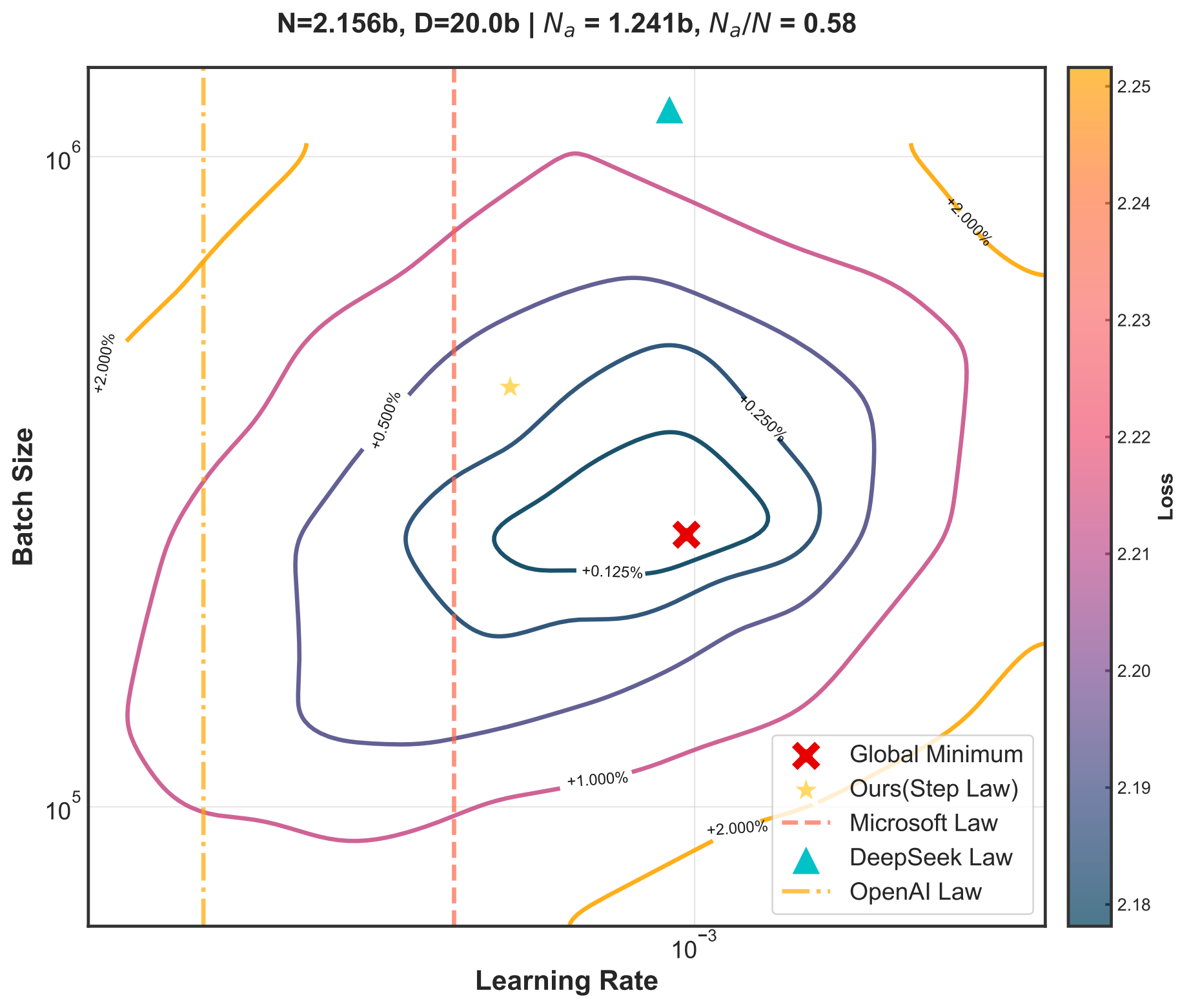
N=2.156b, D=20.0b | N_a = 1.241b, N_a/N = 0.58
Batch Size
Learning Rate
Loss
10^6
10^5
10^{-3}
2.25
2.24
2.23
2.22
2.21
2.20
2.19
2.18
+2.000%
+0.500%
+0.250%
+0.125%
+1.000%
+2.000%
+2.000%
Global Minimum
Ours(Step Law)
Microsoft Law
DeepSeek Law
OpenAI Law